

⚠ Please do not reach out to reviewers directly; messaging them will not speed up your review.

Task Type Taxonomy (Category)

Each task must be labeled with a category from this taxonomy. The category describes the primary theme, topic, or activity in the task.

In addition, there exists another axis of categorization called subtypes, or subcategories (in task.toml)

[Refer to our page on subtypes/subcategories for more information](#)

Categories

system-administration

Tasks involving OS-level configuration, user management, package management, processes, or installing, configuring, and bringing up services, networks, and environments.

Examples:

- Configure a systemd service
- Set up user permissions
- Install and configure Nginx

build-and-dependency-management

Compile code, manage dependencies, build components.

Examples:

- Fix a broken build configuration

- Resolve dependency conflicts
- Set up a multi-stage Docker build

data-processing

Tasks that transform, parse, filter, aggregate datasets or files and directories and generate derived output.

Examples:

- Parse and transform CSV data
- Aggregate log files
- Filter and sort JSON datasets

games

Tasks centered on game-like or simulated environments, interactive puzzles, or simulation games that run in the terminal.

Examples:

- Complete a VimGolf challenge
- Solve a terminal-based puzzle
- Navigate a text adventure

software-engineering

Tasks focused on developing or testing features and algorithms, fixing bugs and improving/optimizing an existing feature, implementing tests, or maintaining software projects.

Examples:

- Implement a caching algorithm
- Fix a race condition
- Optimize database queries

machine-learning

Tasks requiring training, fine-tuning, running inference, or evaluating machine learning models, including dependency setup, running training loops, and managing data pipelines for ML tasks.

Examples:

- Fine-tune a model on custom data
- Debug a training pipeline
- Optimize inference performance

debugging

Tasks that require identifying, diagnosing, and fixing errors in scripts, codebases, or system configurations.

Examples:

- Find and fix a memory leak
- Debug a failing test suite
- Diagnose a production crash

Security

Tasks related to cryptography, authentication, permissions, penetration-style tests, exploit, validate vulnerabilities, reverse engineering or security configuration.

Examples:

- Find a SQL injection vulnerability
- Configure secure TLS settings
- Reverse engineer a binary

scientific-computing

Tasks using scientific libraries or workflows, such as numerical computation, simulations, or domain-specific research code.

Examples:

- Implement a numerical solver
- Debug a simulation
- Optimize a scientific computation

Distribution Guidelines

To ensure benchmark diversity:

- **No single type** should exceed ~30% of total tasks
- **At least four types** should each represent $\geq 10\%$

Choosing a Category

Pick the category that best describes the **primary** activity:

IF THE TASK MAINLY INVOLVES...	USE CATEGORY
OS/server configuration	system-administration
Build systems, packages	build-and-dependency-management
ETL, file processing	data-processing
Interactive challenges	games
Code development, testing	software-engineering
ML model work	machine-learning
Finding/fixing bugs	debugging

IF THE TASK MAINLY INVOLVES...	USE CATEGORY
Security issues	security
Scientific code	scientific-computing

Domain-Specific Tasks

If your tasks are designed for a particular domain (e.g., financial services), an additional taxonomy specific to that domain can be developed. Discuss with the team on Slack.

Next Steps

- [See example tasks](#)
- [Learn difficulty guidelines](#)

PREVIOUS

[Task Requirements](#)

NEXT

[NEW: Submission Diversity Requirements](#)